

KENDRIYA VIDYALAYA SANGATHAN, RAIPUR REGION
FIRST PRE-BOARD EXAM (2024-25)

CLASS – X
SUBJECT – ARTIFICIAL INTELLIGENCE (AI)

M.M. 50
TIME – 2Hr.

General Instruction:

1. Please read the Instruction carefully.
2. This Question paper consists of 21 questions in two section A & section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. Out of the given (5+16=) 21 questions, a candidate has to answer (5+10=) 15 question in the allotted (maximum) time of 2 hours.
5. All questions of a particular section must be attempted in the correct order.
6. SECTION A – OBJECTIVE TYPE QUESTIONS (24 MARKS)
 - i. This section has 05 questions.
 - ii. Marks allotted are mentioned against each question / part.
 - iii. There is no negative marking.
 - iv. Do as per the instructions given.
7. SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS)
 - i. This section has 16 questions.
 - ii. A candidate has to do 10 questions.
 - iii. Do as per the instruction given
 - iv. Marks allotted are mentioned against each question/part.

SECTION – A : OBJECTIVE TYPE QUESTIONS

Q. No.	QUESTIONS	MARKS
Q.1	Answer any 4 out of the given 6 questions on Employability Skills (1x4=4 marks)	
i.	Ramya travelled to Sweden from India to pursue her higher education. But she doesn't know how to speak Swedish (language of Sweden). Because of this, she was unable to find a part time job. This is an example of _____. (a) Interpersonal barrier (b) Physical barrier (c) Organizational barrier (d) Linguistic barrier	1
ii.	Shortcut Key for cut Option. (a) Ctrl + C (b) Ctrl + X (c) Ctrl + K (d) Ctrl + P	1
iii.	Which of the following activities helps in reducing stress? (a) Yoga (b) Walking (c) Meditation (d) All of these	1
iv.	Ritu leaves the company she worked for and starts catering food for marriage programmes. She is an _____. (a) Entrepreneur (b) Employee (c) Cook (d) Unemployed	1
v.	Sonika gets up at 6 am and goes for her hobby classes. Then she comes back home and finishes her homework before going to school. She does all work by herself. No one tells her to do so. This is called _____. (a) Self-Awareness (b) Self-Motivation (c) Self-Regulation (d) Discipline	1
vi.	There are _____ SDGs (a) 13 (b) 17 (c) 12 (d) 15	1
Q.2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	What does the pixel value represent in a grayscale image? (a) The color	1

	(b) The intensity (c) The contrast (d) The brightness	
ii.	How you can identify the problem scoping in the project. (a) Understand why the project was started (b) Define the project's primary objectives (c) Outline the project's work statement. (d) All of the above	1
iii.	Which technique is used to build predictive models? (a) Visual Learning (b) Auditory Learning (c) Machine Learning (d) Reading & writing	1
iv.	What is important to make a Good AI Machine? (a) Algorithm (b) Data (c) Test cases (d) All the above	1
v.	The process of understanding the reliability of any AI model based on output by feeding the test dataset is (a) Data feed (b) Data Reliability (c) Model Evaluation (d) None of these	1
vi.	The higher the value, the more important the word in the document – this is true of which model? (a) Bag of Words (b) TF-IDF (c) YOLO (d) SSD	1
Q.3	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	What is the name of the popular programming language used in data science? (a) Python (b) Java (c) C++ (d) C	1
ii.	AI Machines are composed of (a) Software (b) Hardware (c) Machine/Device (d) Software. Hardware & Machine	1
iii.	_____ element helps us to understand and categorize who is directly and indirectly affected by the problem.	1

	(a) Who (b) What (c) Where (d) Why	
iv.	Which of the following talks about how true the predictions are by any model? (a) Accuracy (b) Reliability (c) Recall (d) F1 Score	1
v.	“Converting text to a common case” is a step in Text Normalisation. (True/False)	1
vi.	Which format is typically used to store a color image in digital form? (a) Grayscale (b) Binary (c) RGB (d) Indexed	1
Q.4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Recall method is not depend on True negative. (True/False)	1
ii.	What does a pixel represent in an image? (a) A segment of a video (b) The smallest unit of an image (c) The color depth of an image (d) The brightness of an image	1
iii.	An organization is using a AI enabled Security surveillance system. Surya works as a Security guard in the Organization. Due to usage of AI enabled security system he has a feeling that he may not be needed in the near future. What is this situation? (a) Data Privacy (b) AI Bias (c) AI Access (d) AI is creating Unemployment	1
iv.	What is data in AI? (a) Facts (b) Instruction (c) Information (d) All of the above	1
v.	What do we call the process of dividing a string into component words? (a) Regression (b) Word Tokenization (c) Classification (d) Clustering	1
vi.	Full form of KNN model. (a) K-negative neighbors (b) K-nearest neighbors	1

	(c) K-national neighbors (d) K-negotiation neighbors	
Q.5	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	In Weather forecasting system, to predict the temperature, rainfall and etc. The type of technology used is _____. (a) Computer Vision (b) Interpersonal Intelligence (c) Natural Language Processing (d) Artificial Intelligence	1
ii.	When the prediction is True and reality is False, that condition is termed as (a) TN (b) TF (c) FP (d) FN	1
iii.	.What is/are the reason(s) for AI Bias in data? (a) Improper assumptions (b) Uneven data collection (c) No testing on Data (d) All the above	1
iv.	TF-IDF in NLP is defined as: (a) Term Frequency and Definite Frequency (b) Term Frequency and Indefinite Frequency (c) Term Frequency and Inverse Document Frequency (d) Term Frequency and Integrated Document Frequency	1
v.	Statement 1: F1 score is evaluated based on precision or recall. Statement 2: When the F1 score is 0, the model accuracy is 100% (a) Both statement1 and statement 2 are correct. (b) Both statement1 and statement 2 are incorrect. (c) Statement 1 is correct but statement 2 is incorrect. (d) Statement 1 is incorrect but statement 2 is correct.	1
vi.	After Lemmatization, the words which we are get after removing the affixes is called (a) Lemmat (b) Lemma (c) Lemmatiz (d) Lemmatiza	1
<u>SECTION – B : SUBJECTIVE TYPE QUESTIONS</u>		
	Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks) Answer each question in 20 – 30 words.	

Q.6	 The symbol shown represents “Turn left”. What type of communication uses such symbols and what is the advantage of using that form of communication?	2													
Q.7	Mention any two symptoms of stress.	2													
Q.8	How to delete files and folders permanently from the recycle bin window?	2													
Q.9	List two qualities of successful Entrepreneurs.	2													
Q.10	How can we become responsible about our own environment?	2													
	Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)														
Q.11	Define Intelligence.	2													
Q.12	What are the features of an Artificial Neural Network?	2													
Q.13	Write the use of Data Science.	2													
Q.14	Define pixel and explain its importance in digital images	2													
Q.15	Explain the difference between Stemming and Lemmatization.	2													
Q.16	Define F1 score with formula.	2													
	Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)														
Q.17	What are the different domains of AI	4													
Q.18	Akhil wants to learn how to scope the problem for an AI Project. Explain him the following: (a) 4W Problem Canvas (b) Problem Statement Template	4													
Q.19	Define supervised and unsupervised learning.	4													
Q.20	Define Text normalization and explain the different steps in involved.	4													
Q.21	A binary classification model has been developed to classify the information spread through social media is as either “Fake ” or “Real ”. The model was tested on a dataset of 300 information, and the resulting confusion matrix is as follows: <table border="1"><tr><th colspan="2" rowspan="2">Confusion Matrix</th><th colspan="2">Reality</th></tr><tr><th>YES</th><th>NO</th></tr><tr><th rowspan="2">Prediction</th><th>YES</th><td>150</td><td>40</td></tr><tr><th>NO</th><td>50</td><td>60</td></tr></table> calculate Accuracy, precision, recall and F1 score.	Confusion Matrix		Reality		YES	NO	Prediction	YES	150	40	NO	50	60	4
Confusion Matrix				Reality											
		YES	NO												
Prediction	YES	150	40												
	NO	50	60												